4. Medicaid Managed Care Open-Contract & MLR Engine — Humana NPV Model (v4.2.0)

Actuarial Overview (New-State Open Contract, 10,000 Starting Members, 10-Year Horizon)

Unlike Medicare Supplement or Individual Dental & Vision, Medicaid Managed Care is modeled as an **open-population state procurement contract** starting with 10,000 members over 10 contract years (120 monthly rows):

- **Continuous Churn & Auto-Assignment:** Existing members disenroll every month due to eligibility redeterminations (MD_Disenroll = 25%/yr), while new members continuously join (MD_NewMembers = 2,800/yr, or 233.3/month) and begin churning the following month.
- **Three Rate Cells:** Population is tracked across **Child**, **Adult**, and **Aged/Blind/Disabled (ABD)** rate cells, each with its own member share, capitation relativity, medical loss-ratio relativity, and disenrollment relativity.
- **State Contract Mechanics:** Incorporates a 2% Quality Withhold (85% earn-back), State-Directed Payments (pass-through), budget-neutral Risk Adjustment with lagged acuity catch-up, an **85% Minimum MLR Floor** (or symmetric risk corridor) settled 6 months after each contract year closes, and a 5-year initial contract term followed by a 75% renewal probability (MD_RenewProb) for Years 6–10.

Visual Open-Contract & Settlement Architecture

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                    MEDICAID MANAGED CARE: 10-YEAR (120-MONTH) CONTRACT ENGINE
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I  [PRE-GO-LIVE: t = 0] ──▶ Outflow: −MD_Startup (Network build, IT systems, state readiness)
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[OPEN-POPULATION RATE-CELL ENGINE (3 Rate Cells: Child, Adult, ABD — Each Month k = 0 ... 119)]
Each cell c has Member Share S_c, Capitation Relativity CR_c, Loss-Ratio Relativity MR_c, and
Disenrollment Relativity QR_c:
• Start Members_c(k) = Members_c(k-1) - Leavers_c(k-1) + (MD_NewMembers ÷ 12) × S_c
• Leavers_c(k) = Members_c(k) × [1 - (1 - MD_Disenroll × QR_c)^(1/12)]
• Member Months (MM) = SUM_c [Members_c - 0.5 × Leavers_c + 0.5 × Joiners_c]
(Steady-state membership settles where Monthly Joiners = Monthly Leavers, ~9,850 members).

[MONTHLY OPERATING CASHFLOWS (All Discounted Mid-Month at v^((k+0.5)/12))]
(+) Gross Capitation:
CapWeighted_MM × [MD_Cap1 × (1 + MD_RateUpd)^(Yr-1)] × RiskAdj_Rev_Factor × Lagged_Acuity_Catchup(Yr)
(+) State-Directed Payments (Pass-Through):
MM × [MD_DP1 × (1 + MD_DPTrend)^(Yr-1)] [Added to BOTH Revenue and Medical Costs]
(-) Quality Withhold Holdback:
State withholds MD_WH (2%) of Gross Capitation each month → Received Revenue = GrossCap × (1 - 2%) + DP
(-) Incurred Medical Costs:
CostWeighted_MM × [MD_Cap1 × MD_MCR1 × (1 + MD_Trend)^(Yr-1) × IF(Yr=1, 1 + MD_AcuityLoad, 1)]
× Plan_Acuity × Acuity_Path(Yr) + Directed_Payments
(-) Premium Tax & Admin:
(GrossCap + DP) × MD_PremTax + MM × [MD_Admin1 × (1 + ExpInfl)^(Yr-1)]

[ANNUAL SETTLEMENTS (Paid 6 Months After Contract Year Closes: t = Yr + 0.5) & RENEWAL WEIGHTING]
(+) Quality Withhold Earn-Back: 2% Withhold × 85% Earn-Back Rate (1.7% of Capitation returned at Yr + 0.5).
(-) Minimum MLR Remittance / Risk Corridor Settlement:
MLR_Base = Received_Revenue + Withhold_Earned_Back
If Floor Rule: Remittance = max(0, 85% × MLR_Base - Medical_Costs) [Paid to state at Yr + 0.5]
(×) Contract Renewal Probability:
Years 1 to 5 (Initial Term): Weight = 100%
Years 6 to 10 (Renewal Term): All cashflows & reserves weighted by MD_RenewProb (75%).
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Step-by-Step Mathematical Equations (Medicaid_Monthly & Medicaid_NPV)

1. Rate-Cell Normalization on the Assumptions Sheet

To guarantee that turning on rate cells (F_RateCells = 1) preserves the exact contract-level Year-1 blended assumptions:

- **Normalized Shares:**
  MD_S_c = MD_Sh_c ÷ (MD_Sh_1 + MD_Sh_2 + MD_Sh_3)
- **Normalized Capitation Relativities:**
  MD_CR_c = MD_CapR_c ÷ (MD_S_1 × MD_CapR_1 + MD_S_2 × MD_CapR_2 + MD_S_3 × MD_CapR_3)
- **Normalized Loss-Ratio Relativities:**
  MD_MR_c = MD_MedR_c ÷ (MD_S_1 × MD_CR_1 × MD_MedR_1 + MD_S_2 × MD_CR_2 × MD_MedR_2 + MD_S_3 × MD_CR_3 × MD_MedR_3)
- **Normalized Churn Relativities:**
  MD_QR_c = MD_ChR_c ÷ (MD_S_1 × MD_ChR_1 + MD_S_2 × MD_ChR_2 + MD_S_3 × MD_ChR_3)

2. Monthly Open-Population Roll-Forward by Rate Cell (Cols D–O)

For each rate cell c in {1: Child, 2: Adult, 3: ABD} and month k in 0..119:

- **Cell Monthly Churn Rate:**
  q_churn_c = 1 – (1 – min(1.0, MD_Disenroll × MD_QR_c))^(1/12)
- **Start-of-Month Cell Membership (Cols E–G):**
  • Month 0: Members_c(0) = MD_Start × MD_S_c
  • Month k: Members_c(k) = Members_c(k – 1) – Leavers_c(k – 1) + (MD_NewMembers ÷ 12) × MD_S_c
- **Monthly Leavers & Joiners (Cols H–L):**
  Leavers_c(k) = Members_c(k) × q_churn_c
  Joiners_c(k) = (MD_NewMembers ÷ 12) × MD_S_c
- **Cell Member Months (Mid-Month Convention):**
  MM_c(k) = Members_c(k) – 0.5 × Leavers_c(k) + 0.5 × Joiners_c(k)
- **Total, Capitation-Weighted, and Cost-Weighted Member Months (Cols M–O):**
  • MM_total(k) = MM_1(k) + MM_2(k) + MM_3(k)
  • MM_cap_wt(k) = MM_1(k) × MD_CR_1 + MM_2(k) × MD_CR_2 + MM_3(k) × MD_CR_3

• MM_cost_wt(k) = MM_1(k) × MD_CR_1 × MD_MR_1 + MM_2(k) × MD_CR_2 × MD_MR_2 + MM_3(k) × MD_CR_3 × MD_MR_3

3. Revenue, Risk Adjustment, Directed Payments & Medical Costs (Cols P–AA)

- **Blended Capitation & Medical PMPM Rates (Cols P–Q):**
  CapPMPM(k) = MD_Cap1 × (1 + MD_RateUpd + MD_RateAdd)^(Year(k) – 1)
  MedPMPM(k) = MD_Cap1 × MD_MCR1 × (1 + MD_Trend + MD_TrendAdd)^(Year(k) – 1) × (1 + MD_Acuity if Year(k) = 1, else 1.0)
- **Risk-Adjusted Revenue & Lagged Acuity Catch-Up:**
  MD_RARev = 1 + MD_RACap × (MD_Acu – 1)
  MD_RAP(Year) = 1 + MD_RACap × (MD_AcuPath(Year – MD_RALag) – 1) if (Year – MD_RALag) ≥ 1, else 1.0
- **Gross Capitation (Col Z):**
  GrossCap(k) = MM_cap_wt(k) × CapPMPM(k) × MD_RARev × MD_RAP(Year(k))
- **State-Directed Payments Pass-Through (Col AA):**
  DP(k) = MM_total(k) × MD_DPe × (1 + MD_DPTrend)^(Year(k) – 1)
- **Revenue Received Net of Quality Withhold (Col R):**
  Revenue(k) = GrossCap(k) × (1 – MD_WHe) + DP(k)
- **Incurred Medical Costs (Col S):**
  Medical(k) = MM_cost_wt(k) × MedPMPM(k) × MD_RAMed × MD_AcuUsePath(Year(k)) + DP(k)
- **Premium Tax & Admin (Cols T–U):**
  PremTax(k) = [GrossCap(k) + DP(k)] × MD_PremTax
  Admin(k) = MM_total(k) × MD_Admin1 × (1 + ExpInfl)^(Year(k) – 1)

4. Annual Quality Withhold Earn-Back & MLR Floor / Corridor Settlement (Medicaid_NPV)

At the end of each contract year y in 1..10, annual totals Rev(y), Med(y), and GrossCap(y) are summed across the 12 months:

- **Quality Withhold Earned Back (Col Z, received at y + 0.5):**
  WithholdEarn(y) = MD_WHe × MD_WHEe × GrossCap(y) **(e.g., 2% withhold × 85% earn-back = 1.7% of Gross Capitation)**
- **MLR Revenue Base (Col AA):**
  MLR_Base(y) = Rev(y) + WithholdEarn(y)
- **State MLR Settlement (Col AB, paid at y + 0.5):**

    - **If Minimum MLR Floor Rule (MD_Rulee = 0, 42 CFR 438.8 standard):**
      MLR_Remit(y) = max(0, MD_MLRFloor × MLR_Base(y) − Med(y))
    - **If Symmetric Risk Corridor Rule (MD_Rulee = 1, Target T, Bands B1 and B2, Plan Shares S2 and S3):**
      Let realized MLR be m(y) = Med(y) ÷ MLR_Base(y).
      GainReturned(y) = MLR_Base(y) × [(1 − S2) × min(max(T − B1 − m(y), 0), B2 − B1) + (1 − S3) × max(T − B2 − m(y), 0)]
      LossShared(y) = MLR_Base(y) × [(1 − S2) × min(max(m(y) − T − B1, 0), B2 − B1) + (1 − S3) × max(m(y) − T − B2, 0)]
      MLR_Remit(y) = GainReturned(y) − LossShared(y)
- **Net Settlement Paid to State (Col G):**
  NetSettle(y) = MLR_Remit(y) − WithholdEarn(y)

Key Valuation Results & The 3 Big Medicaid Insights

Valuation Basis / Sensitivity	Cash NPV	DE NPV (After Tax & Capital)	Actuarial Mechanism & Explanation
Deterministic Base (Uniform 25% Churn Across Cells)	**+$519,615**	**−$1,089,199**	Positive pre-tax cash flow (~1.6% margin), but fails to earn the 8.5% hurdle rate on 15% required RBC capital (ALERT in drift monitor).
Stochastic Monte Carlo Mean (With 85% MLR Floor)	**−$2,074,277**	**−$3,190,107**	**The MLR Floor Trap:** In favorable scenarios where MLR < 85%, Humana remits gains to the state; in adverse scenarios, Humana absorbs 100% of losses. The MLR floor costs **−$2.77M** in expected value (mean Cash

Valuation Basis / Sensitivity	Cash NPV	DE NPV (After Tax & Capital)	Actuarial Mechanism & Explanation
			NPV is +$0.69M without the floor vs. –$2.07M with it).
MACPAC Differential Churn by Rate Cell (Child 22.6% / Adult 33.3% / ABD 10.7%)	**+$5,620,197**	**+$2,710,000**	**Churn by Rate Cell Decides the Sign:** Because ABD members churn at less than half the rate of Adults, the in-force book drifts toward high-capitation ABD members (rising from 11.8% to 22.2% of membership), flipping DE NPV from –$1.09M to **+$2.71M**.

v4.2 Bid Recommendation: Do Not Bid at These Terms

- **Capitation Is Not a Lever:** The state sets capitation and certifies it actuarially sound (42 CFR 438.4, ASOP 49), so there is no premium to solve. Instead the v4.2 pricing solve reports, one lever at a time, what would have to be true to meet P(DE NPV < 0) ≤ 10% (currently 57.0%).
- **Required Levers vs. Public Ranges:** A starting medical cost ratio of 82.6% (PV MLR 83.1%) against public MCO results of 87.1–90.6%, or year-1 admin of $10.07 PMPM against a public $39–$46. Both lie outside every public range, so the recommendation is to negotiate terms, reduce the capital need, or decline.
- **How to Say It in the Room:** "That's a target for care management, admin scale and risk-adjustment accuracy, not a veto." And confirm churn by rate cell with the state data book first: at MACPAC churn by cell, DE NPV turns positive (+$2.71M).